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AGE 418
Sensors and Controls in Precision Agriculture

Lecture 7 Compendium:
Data Collection & The Internet of Things (IoT)

Connecting the Farm: From Sensors to the Cloud

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1 1. The Evolution of Data Collection

In Precision Agriculture, data is the raw material. How we collect that data has evolved significantly over the last three decades, moving from manual labor to automated systems.

1.1 1. Manual Collection (The Clipboard Era)

The agronomist walks the field with a notebook or a clipboard.

- **Process:** Observe crop visually, write down "Wilted" or "Pest Damage", drive back to the office, and type data into Excel.
- **Drawbacks:** Slow, prone to human error, low temporal resolution (only one data point per week or month), and significant lag time between observation and action.

1.2 2. Data Logging (The USB Era)

Sensors are placed in the field, connected to a "Data Logger" (a box with memory).

- **Process:** The logger saves data to an SD card or internal memory. The farmer must physically walk to the device, swap the card or plug in a laptop via USB to download the data.
- **Drawbacks:** The "Sneakernet" problem (walking data around). If a sensor breaks on Monday, the farmer might not know until Friday when they check the device, resulting in lost data.

1.3 3. Wireless Telemetry / IoT (The Connected Era)

Sensors stream data automatically to the internet without human intervention.

- **Process:** Sensor $\rightarrow$ Wireless Transmitter $\rightarrow$ Cloud Server $\rightarrow$ Smartphone App.
- **Benefits:** Real-time alerts (e.g., "Pump Failure!"), analysis of historical trends, and remote control capabilities.

2 2. Defining IoT in Agriculture

Internet of Things (IoT)

The network of physical objects ("things") embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the Internet.

In Agriculture, this is often called **AgIoT** or **Smart Farming**. It is the infrastructure that allows the "Digital Twin" of the farm to exist. It transforms a physical farm into a data stream.

3 3. The 4-Layer IoT Architecture

To design an agricultural system, engineers must understand the four layers of the IoT stack. A failure in any layer results in system failure.

3.1 Layer 1: Perception Layer (The "Senses")

This is the physical hardware located in the field.

- **Sensors:** Collect data from the environment (e.g., Soil Moisture probes, Temperature sensors, GPS trackers).
- **Actuators:** Perform physical actions based on signals (e.g., Solenoid valves opening, relays turning on pumps).
- **End-Nodes:** The physical enclosure containing the microcontroller (e.g., Arduino, ESP32, STM32), the battery, and the sensor interface.

3.2 Layer 2: Network Layer (The "Nerves")

The communication channel that moves data from the End-Node to the Internet.

- **Gateways:** Devices that collect data from local nodes and push it to the web.
- **Protocols:** The language used for transmission (e.g., WiFi, GSM, LoRaWAN).

3.3 Layer 3: Middleware / Cloud Layer (The "Brain")

Where the data is stored, processed, and analyzed.

- **Database:** Storing millions of sensor readings (Time-Series Data).
- **Analytics Engine:** Running algorithms (e.g., "If Soil Moisture drops below 20%, trigger an Alert").
- **APIs:** Interfaces that allow different software programs to talk to each other.

3.4 Layer 4: Application Layer (The "Face")

How the human user interacts with the system.

- **Dashboards:** Visual graphs and maps on a computer screen.
- **Mobile Apps:** Push notifications and simple controls on a smartphone.

4 4. Connectivity: The Rural Challenge

In urban areas, high-speed 4G and WiFi are ubiquitous. In rural Tanzania (e.g., remote areas of Mvomero, Kilolo, or Simanjiro), connectivity is a major engineering constraint. We categorize wireless technologies by **Range** and **Power**.

4.1 1. Short Range (Local Area Networks)

- **Bluetooth (BLE):** Range $\approx$ 10m. Used for configuring sensors with a phone while standing next to them.
- **Zigbee / WiFi:** Range $\approx$ 100m. Used inside greenhouses or processing sheds where power is available and sensors are densely packed.

4.2 2. Cellular (GSM / 4G / LTE)

- **Pros:** Works anywhere there is a mobile tower (Vodacom/Airtel/Tigo). High bandwidth (can send photos and video).
- **Cons: High Power Consumption** (drains batteries quickly). **Recurring Cost** (requires a SIM card and data bundles for every single sensor node).

4.3 3. LPWAN (Low Power Wide Area Network)

This is the **Gold Standard** for Agricultural IoT in remote areas.

- **Examples:** LoRaWAN, Sigfox, NB-IoT.
- **Pros:**
  - **Long Range:** 10km - 15km line-of-sight (ideal for large farms).
  - **Low Power:** Batteries can last 5–10 years on a single charge.
  - **Low Cost:** LoRaWAN uses unlicensed radio spectrum (ISM band). You do not pay monthly airtime fees to a telco if you own the gateway.
- **Cons:** Low bandwidth. Can only send tiny packets of data (numbers), not heavy files like photos or video.

Why LoRaWAN fits Tanzania

LoRaWAN allows a farmer to install ONE Gateway (connected to the internet via one SIM card or Ethernet) that listens to hundreds of sensors spread across a 10km radius. It minimizes data costs and maintenance efforts, making it the most cost-effective solution for large-scale rural sensing.

5 5. Edge Computing vs. Cloud Computing

5.1 Cloud Computing

All raw data is sent to the internet server to be processed.

- *Example:* A weather station sends temperature data every 15 minutes. The cloud server calculates the daily average and Growing Degree Days (GDD).

5.2 Edge Computing

Processing is done **on the device** (the Node) before sending data.

- *Why?* To save data bandwidth, battery life, and reduce latency (reaction time).
- *Example:* A camera on a weeding robot. Streaming HD video to the cloud is impossible in rural areas. The "Edge" computer processes the image locally, identifies a weed, and just sends a simple signal: "Weed Detected" or triggers the sprayer directly.

6 6. Power Management in AgIoT

Unlike a smartphone that you charge every night, an agricultural sensor might be placed in a field and not touched for a year. Power management is a critical engineering task.

6.1 Energy Harvesting

1. **Solar:** Small panels (0.5W to 5W). Essential for gateways or high-traffic sensors. Challenge: Dust, bird droppings, and shading can reduce efficiency. 2. **Lithium Batteries (Li-SOCl2):** High energy density, non-rechargeable. Used in LPWAN nodes to provide 5+ years of life without solar panels.

6.2 Sleep Modes

Engineers program microcontrollers to spend most of their time in "Deep Sleep."

- **Active Mode:** 100 mA consumption (Reading sensor, transmitting). Duration: 2 seconds.
- **Sleep Mode:** 0.01 mA consumption (Waiting). Duration: 59 minutes, 58 seconds.

This duty cycle extends battery life from days to years.

7 7. Wireless Sensor Networks (WSN)

A WSN is a collection of spatially dispersed sensors dedicated to monitoring physical conditions.

7.1 Topologies

- **Star Topology (e.g., LoRaWAN):** Every sensor node talks directly to the central gateway. It is simple, robust, and saves battery because nodes don't have to relay messages for others.
- **Mesh Topology (e.g., Zigbee):** Sensors talk to each other, passing the message along like a bucket brigade until it reaches the gateway.
  - *Pro:* Can get around obstacles (hills, dense forests).
  - *Con:* High battery consumption (nodes must stay awake to listen for neighbors) and higher complexity.

8 8. Lecture Assessment

8.1 Part A: Standard Concept Questions

1. Define **IoT** in the context of agriculture.
2. List the four layers of the IoT Architecture and give one example component for each.
3. Compare **Manual Data Collection** vs. **Automated Telemetry**. List two advantages of automation.

4. Explain the difference between **WiFi** and **LoRaWAN**. Why is LoRaWAN preferred for soil moisture monitoring in a 500-acre farm?
5. What is **Edge Computing**? Why is it necessary for agricultural robotics (e.g., weeding robots)?
6. Define **Latency** and **Bandwidth**. Which IoT protocol has high bandwidth but high power consumption?
7. Describe the function of an **IoT Gateway**.
8. What is a **Sleep Cycle** in sensor electronics, and why is it critical for rural applications?
9. Explain the difference between **Star** and **Mesh** network topologies.
10. Identify two environmental challenges (physical) that IoT hardware faces in a Tanzanian field (e.g., heat, theft) and propose engineering solutions.

8.2 Part B: Advanced Analysis (Open Source Data)

These questions require internet access and the use of open-source datasets/tools.

1. **The Things Network (TTN) Map:** Visit **The Things Network Map** (`https://www.thethingsnetwork.org/map`). *Task:* Check if there are any active LoRaWAN gateways in Tanzania (Zoom into Dar es Salaam, Arusha, or Dodoma). If you were establishing a smart farm in **Morogoro (SUA)**, and there is no gateway, explain the hardware requirements to build one.
2. **Coverage Mapping (RF Planning):** Use the **CloudRF** or **HeyWhatThats** (free web tools) to simulate radio coverage. *Task:* Pick a hill on the SUA farm (e.g., near the weather station). Place a hypothetical LoRa gateway at 10m height. Generate a "Viewshed" or coverage map. Does the signal reach the rice fields in the valley? How does topography affect the signal?
3. **OpenWeatherMap API Integration:** Visit **OpenWeatherMap.org** and sign up for a free API key. *Task:* Read the API documentation for "Current Weather." Write a pseudocode (logic) script that acts as an irrigation controller. *Logic:* Fetch `rain_1h`. IF `rain_1h` > 0, THEN `Turn_Pump_OFF`. Explain how this API integration saves water compared to a simple timer.
4. **Solar Power Budgeting:** You are designing a node with an ESP32 microcontroller (consumes 100mA when active, 0.01mA in sleep) and a cellular modem (consumes 200mA when sending). *Task:* If the device wakes up for 1 minute every hour to send data, calculate the daily power consumption in mAh. Select a battery size (e.g., 18650 Li-Ion 2500mAh) and calculate how many days it will last without a solar panel.
5. **Data Packet Decoding (Payload):** In LoRaWAN, to save bandwidth, we send **Hexadecimal** bytes, not text. *Task:* A sensor sends the payload `01 0B`. *Scenario:* The first byte `01` is the Sensor ID. The second byte `0B` is the Temperature in Celsius (Hexadecimal). Convert `0B` Hex to Decimal. What is the temperature? Why do we send `0B` instead of the text string "11 degrees"?